

Closet Coach :

AI Fashion Assistant for Sustainable Decision-Making

¹Juha Kim, ²Chaeyoung Shim
¹Fashion Business Management, Fashion institute of Technology
²Fashion Business Management, Fashion institute of Technology

INTRODUCTION

The fashion industry produces nearly **10% of global carbon emissions**, yet most consumers have little awareness of how much their own clothes contribute to that number. Numerous brands release sustainability reports, but they remain **distant from everyday shopping decisions**. While consumers understand that reusing or repairing clothes reduces emissions, the challenge is the lack of real-time, item-specific guidance which helps sustainable action develop in a clearer way.

This research introduces **“Closet Coach,”** a fashion assistant app powered by artificial intelligence (AI) built on the Bubble no-code platform. Instead of tracking purchases, Closet Coach allows users to **analyzes their clothes they already own** by inputting each item's category, material, and condition. The system then uses carbon footprint data from multiple credible sources like **Carbonfact** and **Carbonfootprint.com** to calculate the user's emissions and provide actionable insights on the most sustainable next step — reuse, repair, resale, or replace.

Although this application prototype is currently reliant on manual data entry instead of full AI image recognition, it indicates a step for future use of AI tools, like Google Teachable Machine, to classify clothing. The focus of this project is moving consumers from buying more clothing to using existing clothes smarter, which demonstrates how digital tools can turn sustainability into a simple, personal habit.

PROBLEM

1. Awareness Without Action

Despite growing awareness of fashion's environmental cost, consumers rarely act sustainably because they **don't know what difference their choices make**. Brand-level reports fail to translate into personal impact or actionable advice.

2. Lack of Personalized Carbon Data

Most carbon measurement tools (like carbonfootprint.com) focus on transportation or energy use, not fashion consumption. Consumers have **no way to measure the carbon footprint of their own wardrobes** in real time. Even when data exist, they are too technical or generalized to guide daily behavior.

3. AI in Fashion Still Prioritized Consumption

As SeeStone (2024) notes, AI is transforming how consumers *buy* fashion — predicting trends, recommending products, and improve retail sales. However, **AI rarely encourages sustainable ownership or responsible post-purchase behavior**. This imbalance perpetuates overproduction and waste.

Therefore, this research project raises this question :

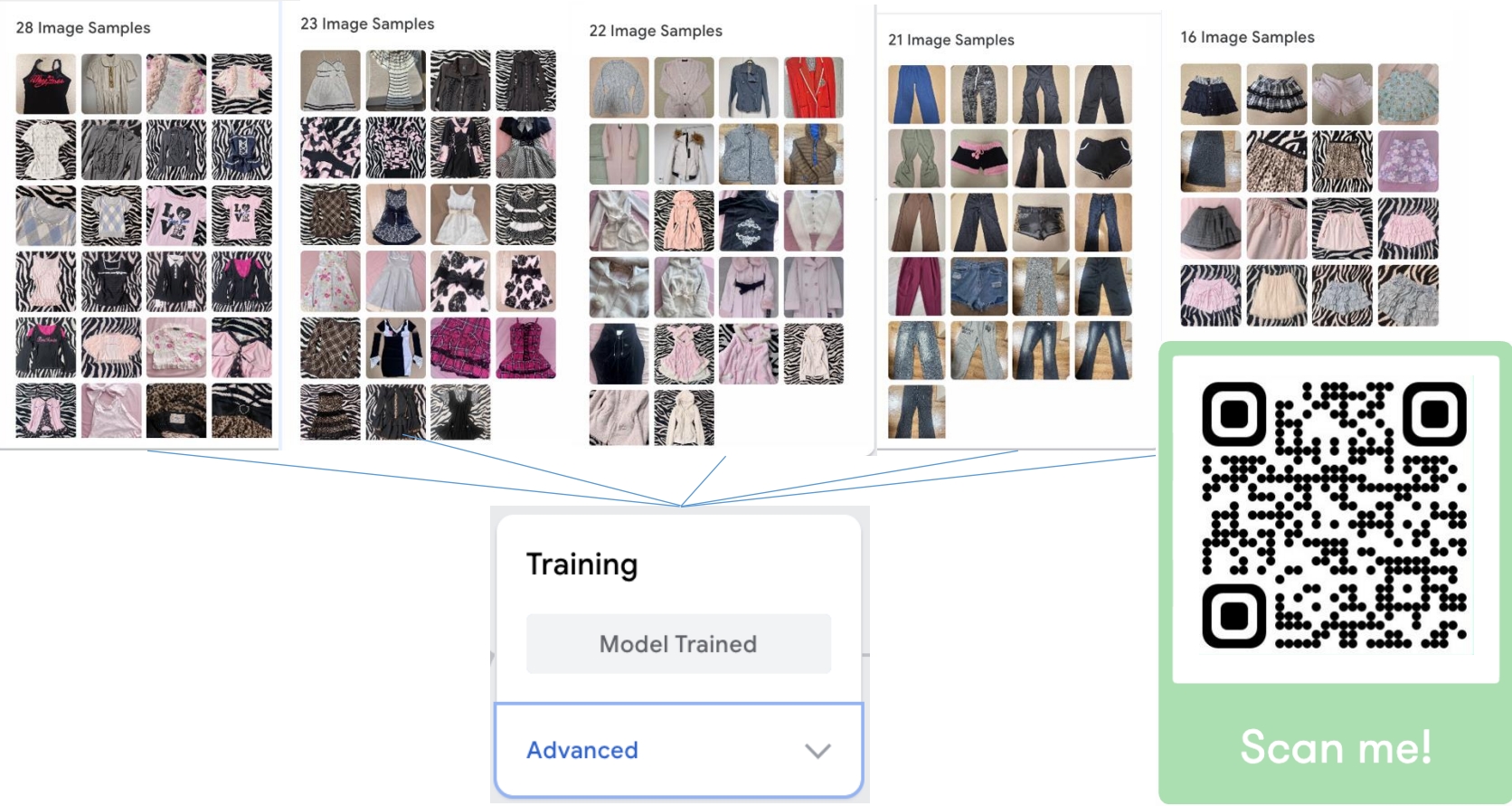
“Can AI transform a user's photo into a personalized recommendation that guides consumers toward the most sustainable next action?”

METHODOLOGY

Step 1. 100 Closet Photos vs. AI : When AI Learns What to Reuse, Repair, or Resell

Using 100 actual photographs directly from our closet, we **trained and tested an AI model**.

The AI learned to classify five areas of clothing — Tops, Skirts, Pants, Outerwear, Dresses. Using **Google Teachable Machine**, we uploaded approximately 20 images for each class/category, and then tested the model on real photographs that it had never seen to assess the model's accuracy. This experiment allowed us to examine the AI's interpretation of fashion images and how accurately it was able to identify real clothes items — bridging data science and fashion's sustainability.



PROPOSED SOLUTIONS & OUTCOMES

Expectations & Results

1. From measurement to Motivation

Closet Coach transforms sustainability from a vague idea into a **personal, data-based experience**.

Instead of abstract numbers, users can receive clear feedback — seeing how their own closet choices impact the environment.

3. Broader Implications

For Consumers: Encourages sustainable decisions through visualized feedback.

For Industry: Shows how no-code tools and AI can help brands create transparent, user-friendly sustainability apps.

For Education: Creates a bridge between AI literacy and fashion sustainability for students and researchers

2. Actionable, Personalized Feedback

Rather than showing general message like “Repairing saves 7kg CO₂e”, Closet Coach gives a dynamic report screen with each item's **carbon footprint** and **recommended action**.

For example:

- Carbon Footprint: 1.4kg CO₂e
- Recommended Action: Replace
- Explanation: “Your garment is no longer wearable or repairable. Replacing it with recycled materials can lower your long-term footprint.”

METHODOLOGY

The *Closet Coach* prototype is a combination of **AI image testing** (using Google Teachable Machine), carbon calculations based on life cycle assessment (**LCA**), and a **behavior simulation** built in Bubble. Although the current version is based on a user-input, it demonstrates how future AI image recognition could use automated garment detection with real time sustainability feedback.

Step 2. What Should You Do With Your Clothes? Explaining the logic of Closet Coach

This step explains the logic behind how *Closet Coach* makes its recommendations.

- The decision engine assesses one of four actions for each garment — **Reuse, Repair, Resell, Replace**.
- In order to obtain an estimate of an item's carbon footprint, users will choose a **category, material, and condition** for the item.
- Each option will be evaluated on three factors: **Carbon (environmental savings), Cost (cost savings), and Convenience (time/work required)**.
- A brief description ("What This Means") would explain how the recommendation helps the user with their sustainability goal.

Option	Description	Estimated Impact
Reuse	Continue wearing for +α uses	Reduces CO ₂ e compared to buying new
Repair	Extend lifespan by small fix	Moderate CO ₂ e reduction at a small cost
Resell	Give or sell it to a new owner	Avoids full production CO ₂ e of new item
Replace	Buy a lower impact alternative	New CO ₂ e, but depends on material choice

Step 3. Can AI Make You a Green Shopper? How to use Closet Coach

This step visualizes how the decision logic was implemented through an interactive no-code prototype.



Conclusion

This project illustrates how Closet Coach can utilize AI, sustainability, and the consumer experience into a no-code platform that is accessible to everyone. By combining LCA data with interactive sliders and dynamic recommendation engines, the prototype is able to visualize how our choices as individuals can impact the environment. Future versions may automate material detection through OCR or image recognition and use APIs from carbon-tracking or resale platforms to create a fully interactive sustainability system.

Limitations & Future Work

- Material detection still relies on user input; future versions may include **label OCR** or **spectral imaging**.
- LCA values vary across regions; updates from **Carbonfact** and **Carbon Trust** will improve accuracy.
- If **resale or repair APIs** were incorporated it could enhance and move this prototype to a fully interactive and sustainable platform.